TRƯỜNG ĐẠI HỌC FPT

AI STUDY HUB

Final Release Document

– Ho Chi Minh, July 2026 –

Table of Contents

I. Deliverable Package

No.	File	Notes
1	SE1904_Group_1_AI_Study_Hub.sql	Database DDL & DML script containing complete database schema, tables, relationships, initial seed data, and demo accounts.
2	SE1904_Group_1_AI Usage Report.xlsx	Detailed AI API token usage log, response latency statistics, model costs, and AI provider performance reports.
3	SE1904_Group_1_SRS.pdf	Software Requirement Document (SRD/SRS) detailing system architecture, function specs, and database design.
4	SE1904_Group_1_Project Tracking.xlsx	Project management tracking spreadsheet detailing task assignments, feature status, complexity, and member progress.
5	SE1904_Group_1_FinalRealease.pdf	Final release document including installation guides, environment configurations, and complete user manual.

Other related deliverables:

- Tagged source codes:
  - Old: https://github.com/nhatlong625/AI_Hub_Study_ver_1
  - New: https://github.com/nhatlong625/AI_Study_Hub_v2
- Demonstration video: https://youtu.be/vZEThGUgIx0

II. Installation Guides

1. Introduction

This section provides detailed instructions for installing, configuring, and running the AI Study Hub system in a local Windows environment.

The AI Study Hub project consists of the following main components:

- **Java Backend:** Spring Boot REST API for authentication, document management, payment, administration, database access, and communication with the AI service.
- **Python Backend:** FastAPI service for AI-powered document summarization, question answering, and practice test generation.
- **Frontend:** React and Vite web application for students, administrators, and public users.
- **Database:** Microsoft SQL Server database containing the system schema and sample data.
- **Cloud Storage:** Supabase Storage or Cloudflare R2 for uploaded document storage.

The default ports used by the system are shown below.

Component	Default Address
Frontend	http://localhost:5173
Java Backend	http://localhost:8080
Python AI Service	http://127.0.0.1:8000
SQL Server	localhost:1433

2. Prerequisites

Before running the project, install the following software.

Software	Required or Recommended Version	Purpose
Windows	Windows 10 or Windows 11	Operating system
Java Development Kit	JDK 25	Running the Spring Boot backend
Apache Maven	Maven 3.9 or later	Installing Java dependencies and running the backend
Node.js	Node.js 18 or later	Running the React frontend
npm	Included with Node.js	Installing frontend dependencies
Python	Python 3.11–3.13; Python 3.12 recommended	Running the FastAPI AI service
Microsoft SQL Server	SQL Server 2019 or later	System database
SQL Server Management Studio	Latest stable version	Importing and managing the database
LibreOffice	Latest stable version, optional	Converting DOCX, PPTX, and other office files to PDF
IntelliJ IDEA	Latest stable version	Opening and running the Java backend
Visual Studio Code	Latest stable version	Opening the frontend and Python backend

Python 3.14 should not be used because some required Python packages may not yet provide stable support for this version.

2.1. Verify the Installed Tools

Open PowerShell and execute the following commands:

```
java --version
mvn --version
node --version
npm --version
python --version
```

The Java version must be JDK 25. If multiple Java versions are installed, configure the JAVA_HOME environment variable to point to the JDK 25 installation directory.

Example:

C:\Program Files\Eclipse Adoptium\jdk-25.0.3.9-hotspot

Restart PowerShell and the IDE after changing the environment variables.

3. Project Structure

After extracting the release package, the main project structure should appear as follows:

AI_Study_Hub_v1/

```
├── backend-java/
|   ├── src/
|   ├── pom.xml
|   └── .env
├── backend-python/
|   ├── src/
|   ├── main.py
|   ├── requirements.txt
|   ├── config.yaml
|   └── .env
├── frontend/
|   ├── src/
|   ├── package.json
|   ├── vite.config.js
|   └── .env
├── AI_Study_Hub.sql
└── README.md
```

The system requires three applications to run simultaneously:

1. Spring Boot Java Backend.
2. FastAPI Python AI Service.
3. React Frontend.

4. Database Installation

4.1. Start SQL Server

Open Windows Services or SQL Server Configuration Manager and make sure the following services are running:

- SQL Server service.
- SQL Server Browser, if a named SQL Server instance is used.

For a local development environment, the default SQL Server port is 1433.

4.2. Enable TCP/IP

If the Java backend cannot connect to SQL Server, enable TCP/IP by completing the following steps:

1. Open **SQL Server Configuration Manager**.
2. Select **SQL Server Network Configuration**.
3. Select **Protocols for the current SQL Server instance**.
4. Right-click **TCP/IP** and select **Enable**.
5. Open the TCP/IP properties.
6. Set the TCP port to 1433 if necessary.
7. Restart the SQL Server service.

4.3. Import the Database

1. Open Microsoft SQL Server Management Studio.
2. Connect to the local SQL Server instance.
3. Select **File → Open → File**.
4. Open the following database script:

AI_Study_Hub.sql

5. Click **Execute**.

The script automatically creates the following database if it does not already exist: AI_StudyHub

Therefore, it is not necessary to manually create the database before executing the script.

4.4. Verify the Database

Execute the following SQL statements:

USE AI_StudyHub;

GO

SELECT * FROM dbo.ROLE;

SELECT * FROM dbo.[USER];

SELECT * FROM dbo.SUBSCRIPTION_PLAN;

The database installation is successful if the tables and sample data are displayed without errors.

The database script also includes two demonstration accounts:

Administrator account:

Email: admin2@aistudyhub.local

Password: Admin@123456

Student account:

Email: student2@aistudyhub.local

Password: Student@123456

These accounts are intended for local demonstration only and must not be used in a production environment.

5. Java Backend Configuration

5.1. Open the Java Backend

Open IntelliJ IDEA and select the following folder:

AI_Study_Hub_v1/backend-java

Wait until IntelliJ IDEA finishes downloading and indexing the Maven dependencies.

Verify that the Project SDK is configured as JDK 25:

1. Select **File → Project Structure**.
2. Open **Project Settings → Project**.
3. Set **Project SDK** to JDK 25.
4. Set the language level to the SDK default or Java 25.

5.2. Create the Java Environment File

Create or update the following file:

backend-java/.env

Use the following configuration as a reference:

```
# Server

SERVER_PORT=8080


# SQL Server

DB_HOST=localhost

DB_INSTANCE=

DB_NAME=AI_StudyHub

DB_USERNAME=sa

DB_PASSWORD=your_sql_server_password

DB_URL=jdbc:sqlserver://localhost:1433;databaseName=AI_StudyHub;encrypt=false;trustServerCertificate=true


# JWT authentication

JWT_SECRET=replace_with_a_secret_key_of_at_least_32_characters
```

```
JWT_EXPIRATION_MS=86400000
JWT_REFRESH_MS=604800000

# Gmail SMTP
MAIL_HOST=smtp.gmail.com
MAIL_PORT=587
MAIL_USERNAME=your_email@gmail.com
MAIL_PASSWORD=your_gmail_application_password

# Frontend addresses
FRONTEND_URL=http://localhost:5173
APP_FRONTEND_URL=http://localhost:5173
EMAIL_VERIFY_URL=http://localhost:5173/verify-email
PASSWORD_RESET_URL=http://localhost:5173/reset-password

# Python AI service
PYTHON_AI_BASE_URL=http://127.0.0.1:8000
PYTHON_INTERNAL_API_KEY=replace_with_the_same_internal_key_used_by_python

# AI API key encryption
AI_CONFIG_MASTER_KEY=replace_with_a_secret_key_of_at_least_32_characters
AI_CONFIG_LEGACY_MASTER_KEY=

# Default storage provider
STORAGE_PROVIDER=SUPABASE

# Supabase Storage
SUPABASE_URL=https://your-project.supabase.co
SUPABASE_SECRET_KEY=your_supabase_service_role_key
SUPABASE_BUCKET=documents
```

Cloudflare R2, required only when STORAGE_PROVIDER=R2

CLOUDFLARE_R2_ACCOUNT_ID=

CLOUDFLARE_R2_ACCESS_KEY_ID=

CLOUDFLARE_R2_SECRET_ACCESS_KEY=

CLOUDFLARE_R2_BUCKET=documents

CLOUDFLARE_R2_PUBLIC_URL=

Google Login

GOOGLE_CLIENT_ID=your_google_oauth_client_id

PayOS

PAYOS_CLIENT_ID=your_payos_client_id

PAYOS_API_KEY=your_payos_api_key

PAYOS_CHECKSUM_KEY=your_payos_checksum_key

LibreOffice, required for converting Office documents to PDF

DOCUMENT_CONVERSION_OFFICE_HOME=C:/Program Files/LibreOffice

Important configuration notes:

- JWT_SECRET must contain at least 32 characters.
- AI_CONFIG_MASTER_KEY must contain at least 32 characters.
- PYTHON_INTERNAL_API_KEY must have exactly the same value in the Java and Python .env files.
- Do not include spaces before or after the = character.
- Do not commit the real .env file to GitHub.
- Do not publish database passwords, Gmail passwords, storage keys, PayOS keys, or AI API keys.

5.3. Gmail Configuration

Email verification and password reset functions use Gmail SMTP.

To configure Gmail:

1. Log in to the Google account used by the application.
2. Enable two-step verification.
3. Create a Gmail application password.
4. Enter the Gmail address into MAIL_USERNAME.
5. Enter the generated application password into MAIL_PASSWORD.

The normal Gmail account password should not be used.

5.4. Storage Configuration

The system supports two storage providers.

Option A: Supabase Storage

Create a Supabase project and a bucket named:

documents

Then configure:

STORAGE_PROVIDER=SUPABASE

SUPABASE_URL=https://your-project.supabase.co

SUPABASE_SECRET_KEY=your_supabase_service_role_key

SUPABASE_BUCKET=documents

The bucket must allow the application to upload and retrieve files. A public bucket or an appropriate signed-URL policy is required for document preview.

Option B: Cloudflare R2

Create a Cloudflare R2 bucket and configure:

STORAGE_PROVIDER=R2

CLOUDFLARE_R2_ACCOUNT_ID=your_account_id

CLOUDFLARE_R2_ACCESS_KEY_ID=your_access_key

CLOUDFLARE_R2_SECRET_ACCESS_KEY=your_secret_key

CLOUDFLARE_R2_BUCKET=documents

CLOUDFLARE_R2_PUBLIC_URL=your_public_bucket_url

Only one provider needs to be selected as the default storage provider.

5.5. Google Login Configuration

Create a Google OAuth Web Application credential and add the following authorized JavaScript origin:

http://localhost:5173

Copy the generated Client ID into both files:

```
# backend-java/.env
GOOGLE_CLIENT_ID=your_google_oauth_client_id

# frontend/.env
VITE_GOOGLE_CLIENT_ID=your_google_oauth_client_id
```

Both values must be identical.

5.6. PayOS Configuration

To enable online subscription payment, create a PayOS merchant account and enter the provided credentials:

```
PAYOS_CLIENT_ID=your_payos_client_id
PAYOS_API_KEY=your_payos_api_key
PAYOS_CHECKSUM_KEY=your_payos_checksum_key
```

If these values are not configured, the remaining system functions can still run, but PayOS payment-link creation will not be available.

5.7. Install Java Dependencies

Open PowerShell in the backend-java directory:

```
cd backend-java
mvn dependency:resolve
```

Alternatively, Maven dependencies can be downloaded by executing:

```
mvn clean package -DskipTests
```

5.8. Run the Java Backend

Run the following command:

```
mvn spring-boot:run
```

Alternatively, open the following class in IntelliJ IDEA and click **Run**:

src/main/java/com/aistudyhub/AiStudyHubApplication.java

The backend is running successfully when the console displays that the application has started on port 8080.

Verify the service by opening:

http://localhost:8080/api/health

6. Python AI Service Configuration

6.1. Open the Python Backend

Open a new PowerShell window and navigate to:

cd backend-python

6.2. Create a Virtual Environment

Run:

py -3.12 -m venv .venv

Activate the virtual environment:

.\.venv\Scripts\Activate.ps1

After activation, the terminal should display (.venv) before the command prompt.

If PowerShell prevents script execution, run:

Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass

Then activate the virtual environment again.

6.3. Install Python Dependencies

Run:

python -m pip install --upgrade pip

pip install -r requirements.txt

The main Python dependencies include:

- FastAPI.
- Uvicorn.
- Pydantic.
- Google Generative AI.
- HTTPX.
- Python Dotenv.

6.4. Create the Python Environment File

Create or update:

backend-python/.env

Use the following configuration:

```
# AI provider selection

LLM_PROVIDER=gemini


# Gemini

GEMINI_API_KEY=your_gemini_api_key

GEMINI_MODEL=gemini-2.5-flash


# OpenAI, optional

OPENAI_API_KEY=

OPENAI_MODEL=gpt-4o-mini


# DeepSeek, optional

DEEPSEEK_API_KEY=

DEEPSEEK_MODEL=deepseek-chat


# Internal Java–Python authentication

PYTHON_INTERNAL_API_KEY=replace_with_the_same_internal_key_used_by_java
```

Allowed frontend origins

PYTHON_CORS_ORIGINS=http://localhost:5173,http://127.0.0.1:5173

At least one valid AI provider API key should be configured for real AI responses.

Supported values for LLM_PROVIDER include:

auto

gemini

openai

deepseek

When auto is selected, the service selects an available configured AI provider according to the application logic.

6.5. Obtain a Gemini API Key

To use Gemini:

1. Access Google AI Studio.
2. Log in using a Google account.
3. Select the option to create an API key.
4. Copy the generated API key.
5. Paste the key into GEMINI_API_KEY.

The API key must never be included in screenshots, public repositories, or submitted documentation.

6.6. Run the Python AI Service

Execute:

```
python -m uvicorn main:app --reload --host 127.0.0.1 --port 8000
```

The AI service is running successfully when Uvicorn displays:

```
Uvicorn running on http://127.0.0.1:8000
```

Verify the service using:

```
http://127.0.0.1:8000/health
```

The FastAPI documentation is available at:

```
http://127.0.0.1:8000/docs
```

7. Frontend Configuration

7.1. Open the Frontend

Open a third PowerShell window and navigate to:

cd frontend

7.2. Create the Frontend Environment File

Create or update:

frontend/.env

Add the following configuration:

VITE_API_BASE_URL=http://localhost:8080/api

VITE_API_URL=http://localhost:8080/api

VITE_AI_STUDY_USER_ID=1

VITE_GOOGLE_CLIENT_ID=your_google_oauth_client_id

The value of VITE_GOOGLE_CLIENT_ID must match GOOGLE_CLIENT_ID in the Java backend configuration.

7.3. Install Frontend Dependencies

Run:

npm install

The main frontend technologies include:

- React 18.
- React Router.
- Vite.
- Tailwind CSS.
- Supabase JavaScript libraries.

7.4. Run the Frontend

Execute:

npm run dev

The frontend is available at:

http://localhost:5173

Open this address using Google Chrome, Microsoft Edge, or another modern browser.

7.5. Verify the Production Build

To verify that the frontend can be built successfully, run:

npm run build

The generated production files will be stored in:

frontend/dist

8. Recommended Startup Order

The system should be started in the following order.

Step 1: Start SQL Server

Ensure that SQL Server is running and that the AI_StudyHub database has been imported.

Step 2: Start the Python AI Service

cd backend-python

.\.venv\Scripts\Activate.ps1

python -m uvicorn main:app --reload --host 127.0.0.1 --port 8000

Step 3: Start the Java Backend

cd backend-java

mvn spring-boot:run

Step 4: Start the React Frontend

cd frontend

npm run dev

Step 5: Open the Application

Open:

http://localhost:5173

9. Installation Verification

The installation is successful when all of the following checks pass.

No.	Check
1	Open http://127.0.0.1:8000/health
2	Open http://127.0.0.1:8000/docs
3	Open http://localhost:8080/api/health
4	Open http://localhost:5173
5	Log in using a demonstration account
6	Upload a PDF document
7	Ask a question about a document

8	Generate a practice test
9	Open the pricing page
10	Create a PayOS payment link

10. Common Installation Problems

10.1. Java Uses the Wrong JDK Version

Problem:

The configured JDK does not support the required JVM target.

Solution:

- Install JDK 25.
- Set JAVA_HOME to JDK 25.
- Configure the IntelliJ Project SDK to JDK 25.
- Restart IntelliJ IDEA and PowerShell.

Verify using:

java --version

mvn --version

10.2. Java Backend Cannot Connect to SQL Server

Check the following items:

- SQL Server is running.
- The database name is AI_StudyHub.
- TCP/IP is enabled.
- Port 1433 is available.
- DB_USERNAME and DB_PASSWORD are correct.
- The DB_URL value is correct.

Example:

DB_URL=jdbc:sqlserver://localhost:1433;databaseName=AI_StudyHub;encrypt=false;trustServerCertificate=true

10.3. Port 8080 Is Already in Use

Find the process using port 8080:

netstat -ano | findstr :8080

Stop the process:

Stop-Process -Id <PID> -Force

10.4. Port 8000 Is Already in Use

Run:

netstat -ano | findstr :8000

Stop-Process -Id <PID> -Force

10.5. Port 5173 Is Already in Use

Run:

netstat -ano | findstr :5173

Stop-Process -Id <PID> -Force

Then restart the frontend.

10.6. Frontend Cannot Call the Backend

Check the frontend environment variables:

VITE_API_BASE_URL=http://localhost:8080/api

VITE_API_URL=http://localhost:8080/api

After modifying .env, stop and restart:

npm run dev

10.7. AI Returns a Mock Response or Service-Unavailable Error

Check the following items:

- The Python service is running on port 8000.
- PYTHON_AI_BASE_URL is correct.
- PYTHON_INTERNAL_API_KEY is identical in both backend environment files.
- At least one AI API key is valid.
- The AI account has remaining quota.
- The selected AI model is available.

10.8. Invalid Internal API Key

Problem:

401 Invalid internal API key

Solution:

Use the same value in both files:

backend-java/.env

PYTHON_INTERNAL_API_KEY=a_shared_secret_value

backend-python/.env

PYTHON_INTERNAL_API_KEY=a_shared_secret_value

Restart both services after changing the value.

10.9. Document Upload Fails

Check the selected storage provider.

For Supabase:

- The documents bucket exists.
- The Supabase URL is correct.
- The service-role key is valid.
- The bucket policy allows file access.

For Cloudflare R2:

- The account ID is correct.
- The access key and secret key are valid.
- The R2 bucket exists.
- The public URL is configured when public document preview is required.

10.10. Office Document Conversion Fails

The system converts office documents to PDF before storing them.

Install LibreOffice and configure:

DOCUMENT_CONVERSION_OFFICE_HOME=C:/Program Files/LibreOffice

Restart the Java backend after changing the configuration.

PDF files do not require LibreOffice conversion.

10.11. Verification or Password-Reset Email Is Not Received

Check:

- MAIL_USERNAME is correct.
- MAIL_PASSWORD is a Gmail application password.
- Gmail two-step verification is enabled.
- Port 587 is not blocked.
- The email is not in the spam folder.
- The Java backend log does not show an SMTP authentication error.

10.12. Google Login Reports an Invalid Client

Check:

- A Web Application OAuth client was created.
- http://localhost:5173 is included in the authorized JavaScript origins.
- The same Client ID is used in both Java and frontend .env files.
- The frontend was restarted after changing .env.

11. Security Notes

The following sensitive information must never be included in public source code, screenshots, reports, videos, or GitHub repositories:

- SQL Server passwords.
- Gmail application passwords.
- JWT secrets.
- AI API keys.
- Supabase service-role keys.
- Cloudflare R2 secret keys.
- PayOS credentials.
- Google OAuth secrets.
- Internal Java–Python API keys.
- AI configuration master keys.

Before submitting the project, replace real values with placeholders and provide private credentials separately to the evaluator when necessary.

The .env files should be excluded through .gitignore.

12. Release Package Cleanup

The following generated folders should not be included in the final source-code ZIP because they can be recreated:

backend-java/target/

backend-python/.venv/

frontend/node_modules/

frontend/dist/

*.log

Dependencies can be restored using:

Java dependencies

cd backend-java

mvn dependency:resolve

Python dependencies

cd backend-python

py -3.12 -m venv .venv

.\.venv\Scripts\Activate.ps1

pip install -r requirements.txt

Frontend dependencies

cd frontend

npm install

Removing generated folders reduces the size of the release package and prevents environment-specific files from causing installation problems on another computer.

III. User Manual

1. Overview

AI Study Hub is an intelligent learning platform designed to help FPT University students store, organize, and study academic materials efficiently. Instead of storing files scattered across multiple cloud storage drives or disorganized personal folders, students can upload all their study materials to a centralized system and neatly organize them by semester and course code.

A highlight of AI Study Hub is its built-in AI learning assistant. When users read uploaded files – including PDFs, Word documents (DOC/DOCX), and presentation slides (PPTX) – directly on the website, they can interact with the AI to ask questions, summarize lecture slides, or clarify difficult concepts. The AI provides accurate answers along with clickable source links, allowing students to click and highlight the exact text or slide in the original document where the answer is found. Additionally, students can use this platform to automatically generate multiple-choice tests from their notes and slides, helping them review knowledge and prepare for exams effectively within a single application.

2. User Workflow

2.1. Workflow: User Registration and Login Workflow

a. Purpose & Actor

- **Purpose:** Enables new users to create an AI Study Hub account and allows existing users to securely log in using email and password or a supported third-party account.
- **Actor:** Guest / Registered User.

b. Step-by-Step Procedure

Part 1: User Registration

- **Step 1:** Guest opens the AI Study Hub landing page and clicks **“Get started”** or **“Create account.”**
- **Step 2:** The system displays the **Create Your Account** page.
- **Step 3:** Guest enters the required registration information:
    - Full Name
    - Email
    - Password

- **Step 4:** Guest reviews and selects the checkbox to agree to the **Terms of Service** and **Privacy Policy**.
- **Step 5:** The system validates the entered information.
- **Step 6:** If any information is invalid, the system displays an appropriate error message, such as:
    - Full name is required.
    - Email format is invalid.
    - Email already exists.
    - Password must contain at least eight characters.
    - Terms and conditions must be accepted.
- **Step 7:** When all information is valid, the **"Create account"** button becomes available.
- **Step 8:** Guest clicks **"Create account."**
- **Step 9:** The system creates the user account and may send a verification email to the registered email address.
- **Step 10:** After successful registration or email verification, the system redirects the user to the Login page or directly to the user dashboard, depending on the configured authentication flow.

Part 2: Third-Party Registration

- **Step 11:** Guest may choose **"Continue with Google"** or **"Continue with Apple."**
- **Step 12:** The system redirects the user to the selected authentication provider.
- **Step 13:** Guest selects an account and grants the required permissions.
- **Step 14:** The authentication provider returns the verified account information to AI Study Hub.
- **Step 15:** If the account does not already exist, the system creates a new user account using the verified information.
- **Step 16:** The system authenticates the user and redirects them to the appropriate dashboard.

Part 3: User Login

- **Step 17:** Registered User opens the AI Study Hub Login page.
- **Step 18:** The system displays the login form with:
    - Email
    - Password
    - Remember Me
    - Forgot Password
    - Google Login
    - Apple Login
- **Step 19:** User enters the registered email address and password.
- **Step 20:** User may select **"Remember me"** to maintain the login session on the current device.
- **Step 21:** User clicks **"Sign In."**
- **Step 22:** The system validates the login credentials and account status.
- **Step 23:** If the credentials are invalid, the system displays an error message such as: Invalid email or password.
- **Step 24:** If the account has not been verified, suspended, or deactivated, the system displays the corresponding notification and prevents access.
- **Step 25:** If the credentials and account status are valid, the system creates an authenticated session and redirects the user to the main dashboard.

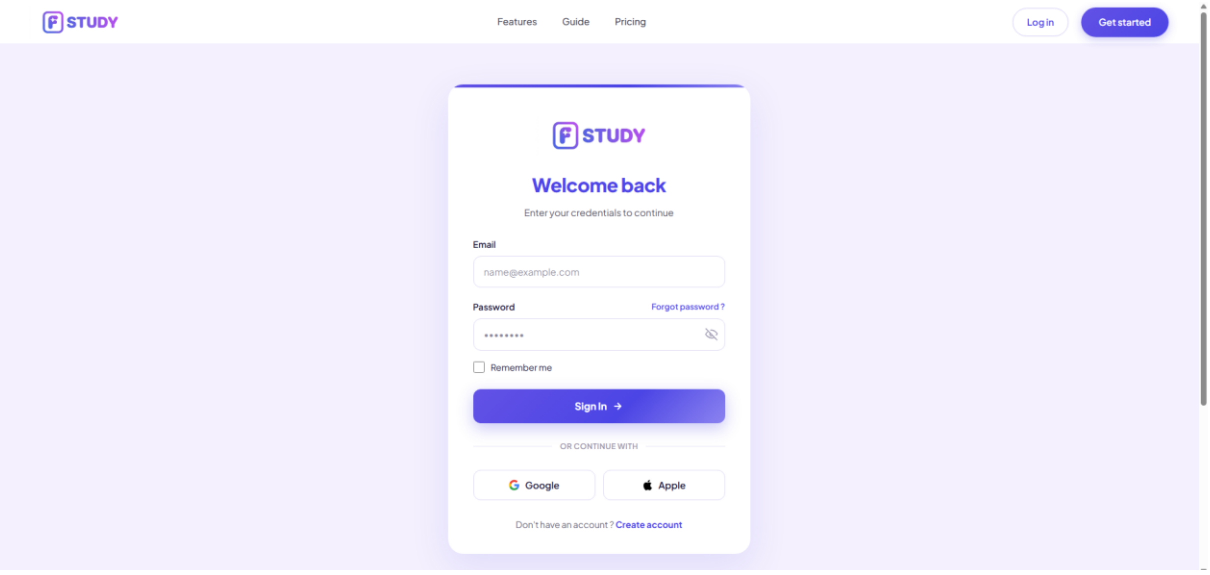
Part 4: Third-Party Login

- **Step 26:** User clicks **“Google”** or **“Apple”** on the Login page.
- **Step 27:** The system redirects the user to the selected authentication provider.
- **Step 28:** User selects an account and completes authentication.
- **Step 29:** The system verifies the returned authentication information.
- **Step 30:** If authentication is successful, the system logs the user in and redirects them to the dashboard.
- **Step 31:** If authentication fails, the system displays an error message and keeps the user on the Login page.

Part 5: Forgot Password Navigation

- **Step 32:** User clicks **“Forgot password?”** on the Login page.
- **Step 33:** The system redirects the user to the password recovery page.
- **Step 34:** User enters the registered email address and requests a password reset link.
- **Step 35:** The system sends password-reset instructions to the registered email address.

c. Screenshots



2.2. Workflow: Subscription Plan Upgrade Workflow

a. Purpose & Actor

- **Purpose:** Enables registered users to review available subscription plans, compare plan benefits, select a billing cycle, upgrade their current subscription, and complete payment to access additional AI Study Hub features.
- **Actor:** Registered User / Student.

b. Step-by-Step Procedure

Part 1: Open Subscription Plan Selection

- **Step 1: User logs in to the AI Study Hub system.**
- **Step 2: The system redirects the user to the Home page.**
- **Step 3: The system displays the user’s current subscription plan in the profile area, such as Basic.**
- **Step 4: User clicks the “Upgrade” button located at the bottom of the sidebar.**

- **Step 5: The system opens the Elevate Your Learning subscription plan window.**
- **Step 6: The system displays the available subscription plans:**
  - **Basic**
  - **Plus**
  - **Pro**
- **Step 7: The system marks the user's existing package as Current Plan.**

Part 2: Review and Compare Subscription Plans

- **Step 8: User reviews the price and included benefits of each subscription plan.**
- **Step 9: The Basic plan displays the currently available free features, such as:**
  - **1 GB cloud storage**
  - **Free starter usage**
  - **Limited support**
  - **Limited access to advanced AI models**
- **Step 10: The Plus plan displays upgraded benefits, such as:**
  - **30 quiz generations per month**
  - **10 GB cloud storage**
  - **Priority email support**
  - **Smart citation generator**
- **Step 11: The Pro plan displays premium benefits, such as:**
  - **Unlimited quiz generations**
  - **50 GB cloud storage**
  - **Advanced AI models**
  - **Offline mode and synchronization**
  - **Dedicated support**
- **Step 12: User selects the preferred billing cycle:**
  - **Monthly**
  - **Yearly**
- **Step 13: The system updates the displayed subscription prices according to the selected billing cycle.**

Part 3: Select an Upgrade Plan

- **Step 14: User clicks "Upgrade to Plus" or "Upgrade to Pro."**
- **Step 15: The system verifies whether the selected plan is different from the user's current plan.**
- **Step 16: If the selected plan is the current plan, the system does not allow the user to continue with an unnecessary upgrade.**
- **Step 17: If the selected plan is valid, the system displays an order summary containing:**
  - **Selected subscription plan**
  - **Billing cycle**
  - **Subscription price**
  - **Subscription duration**
  - **Total payment amount**
- **Step 18: User reviews the subscription information before continuing.**

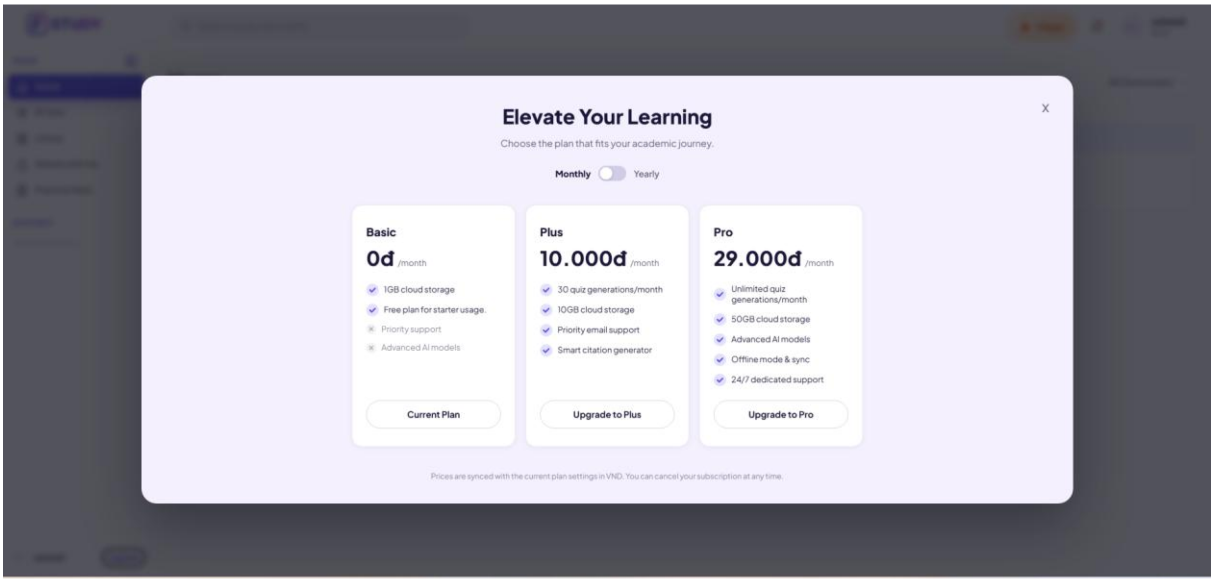
Part 4: Payment Processing

- **Step 19: User clicks “Continue to Payment” or the equivalent confirmation button.**
- **Step 20: The system creates a pending subscription order.**
- **Step 21: The system redirects the user to the configured payment gateway.**
- **Step 22: User selects a supported payment method and completes the payment.**
- **Step 23: The payment gateway processes the transaction and returns the payment result to AI Study Hub.**
- **Step 24: If the payment fails or is cancelled:**
  - **The system keeps the current subscription unchanged.**
  - **The order is marked as failed or cancelled.**
  - **The system displays an appropriate payment error message.**
- **Step 25: If the payment is successful:**
  - **The order is marked as paid.**
  - **The new subscription is activated.**
  - **The user’s available storage and feature quotas are updated.**
  - **The subscription start and expiration dates are recorded.**

Part 5: Upgrade Confirmation

- **Step 26: The system redirects the user back to AI Study Hub.**
- **Step 27: The system displays a successful upgrade notification.**
- **Step 28: The user profile is updated to show the new subscription plan, such as Plus or Pro.**
- **Step 29: The user can immediately access the features and quotas included in the upgraded subscription plan.**
- **Step 30: The system records the subscription and payment information in the user’s transaction history.**

c. Screenshots & UI References



2.3. Workflow: Upload a Document

a. Purpose & Actor

- **Purpose:** Allow a student to add a study document to one of their courses. The uploaded file is validated, converted to PDF, stored, and automatically processed by the AI service to generate a summary.
- **Actor:** Logged-in Student (document owner).

b. Step-by-Step Procedure

- **Step 1:** From the sidebar, open Library, then click a course to open its detail page.
- **Step 2:** Click the **"Upload Document"** button at the top-right of the course page. The system opens the Upload Document dialog.
- **Step 3:** Drag and drop a file into the drop zone, or click **"click to browse"** to choose one. Accepted types: PDF, DOCX, PPTX, PNG, JPG (max 50MB).
- **Step 4:** Enter a **Document Title**. If left blank, the file name is used as the default title.
- **Step 5:** Click the **"Upload"** button. While the file is being processed, the button shows "Uploading…" and is disabled.

***Expected Result:** On success, the document is added to the course as Private, and the system opens the Document Viewer for the new file. AI summarization then runs in the background.*

Validation & Error Cases

- Missing input: if no file, title, or course is provided, the system shows "Please choose a file, title, and course before uploading."
- Storage limit: if the upload would exceed the plan quota, the system shows a storage-limit notice (with used / limit / new-file sizes) and prompts to upgrade.
- Conversion failure: if a non-PDF file cannot be converted to PDF, the upload is rejected with a conversion-failed message.

c. Screenshots & UI References

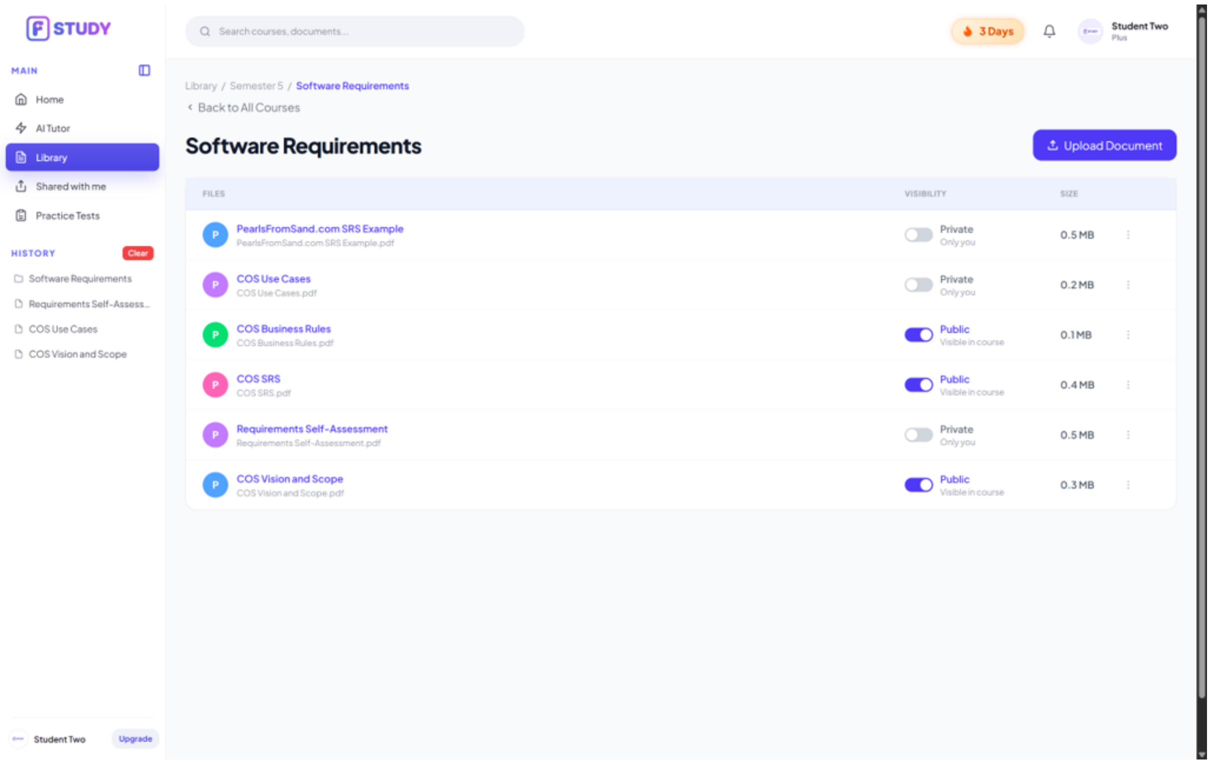


Figure 2.1.1 – Library Course Detail page

● (1) “Upload Document” button (top-right

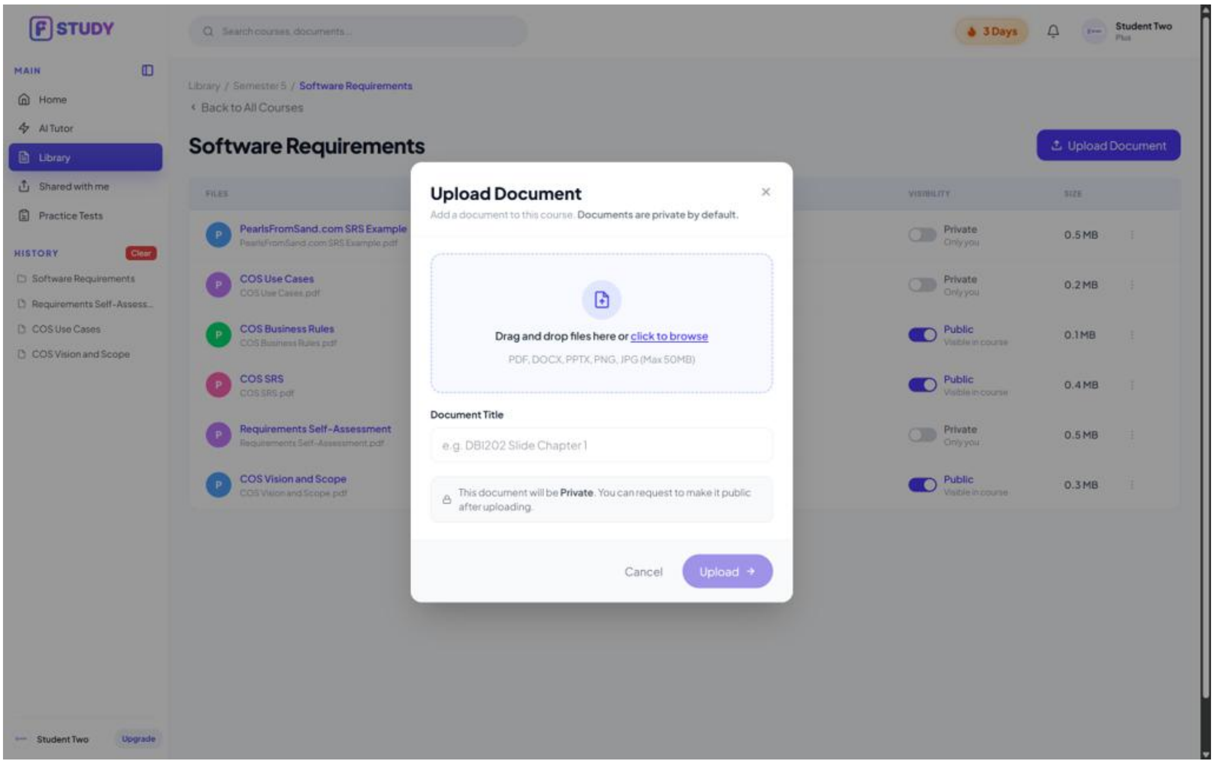


Figure 2.1.2 – Upload Document dialog

● (1) Drag-and-drop zone / “click to browse” link.
● (2) Document Title field.
● (3) “Upload” button.

2.4. Workflow: Manage Documents in a Course

a. Purpose & Actor

- **Purpose:** Allow the owner to manage the documents inside a course: preview a file, rename it, control its visibility (request publication to the public library), and delete it.
- **Actor:** Logged-in Student (document owner). Publishing a document to the public library additionally requires Admin approval.

b. Step-by-Step Procedure

Part 1: Preview, Rename & Delete

- **Step 1:** Open Library and click a course to view its documents. Each document row shows its title, visibility, and size.
- **Step 2:** Click the row action menu (⋮) on a document to open its options: Preview, Edit, Share, Delete.
- **Step 3:** Click **Preview** to open a quick in-page preview of the file in a modal, then close it to return.
- **Step 4:** Click **Edit** to rename the document; enter the new title and save. The document's display title is updated.
- **Step 5:** Click **Delete** to remove the document; confirm in the “Delete document?” dialog.

Expected Result: *The selected action is applied — the preview closes, the title updates, or the document is removed from the course.*

Part 2: Change Visibility (Request Public)

- **Step 6:** On a document row, use the **Visibility toggle**. A Private document shows “Private / Only you” with the hint “Click to request Public”.
- **Step 7:** Turn the toggle on to request publication. The document moves to a pending state (“Awaiting approval”) and is submitted for Admin review.
- **Step 8:** After the Admin approves, the document becomes Public (“Visible in course”) and appears in the public library. The owner may toggle it back to Private.

Expected Result: *Visibility changes to Private → Awaiting approval → Public as the request is submitted and approved.*

c. Screenshots & UI References

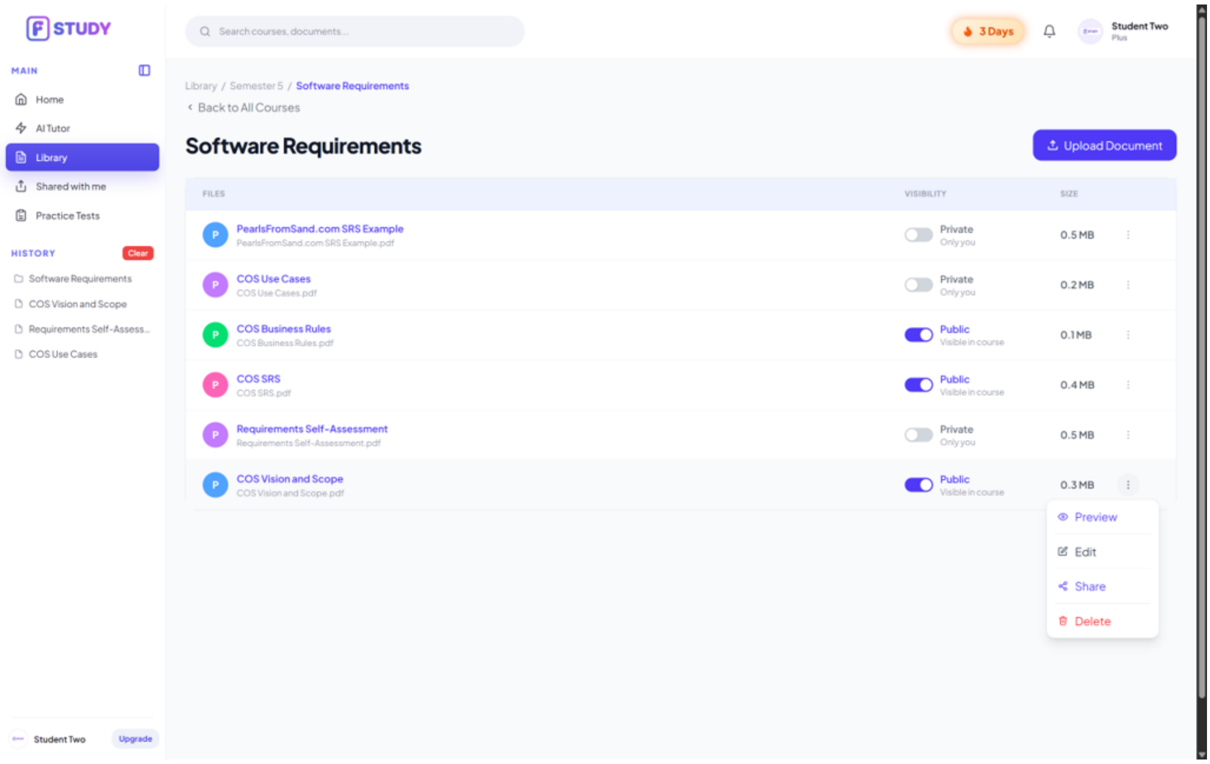


Figure 2.2.1 – Document row action menu

- Row action menu (⋮) trigger.
- Preview / Edit / Share / Delete options.

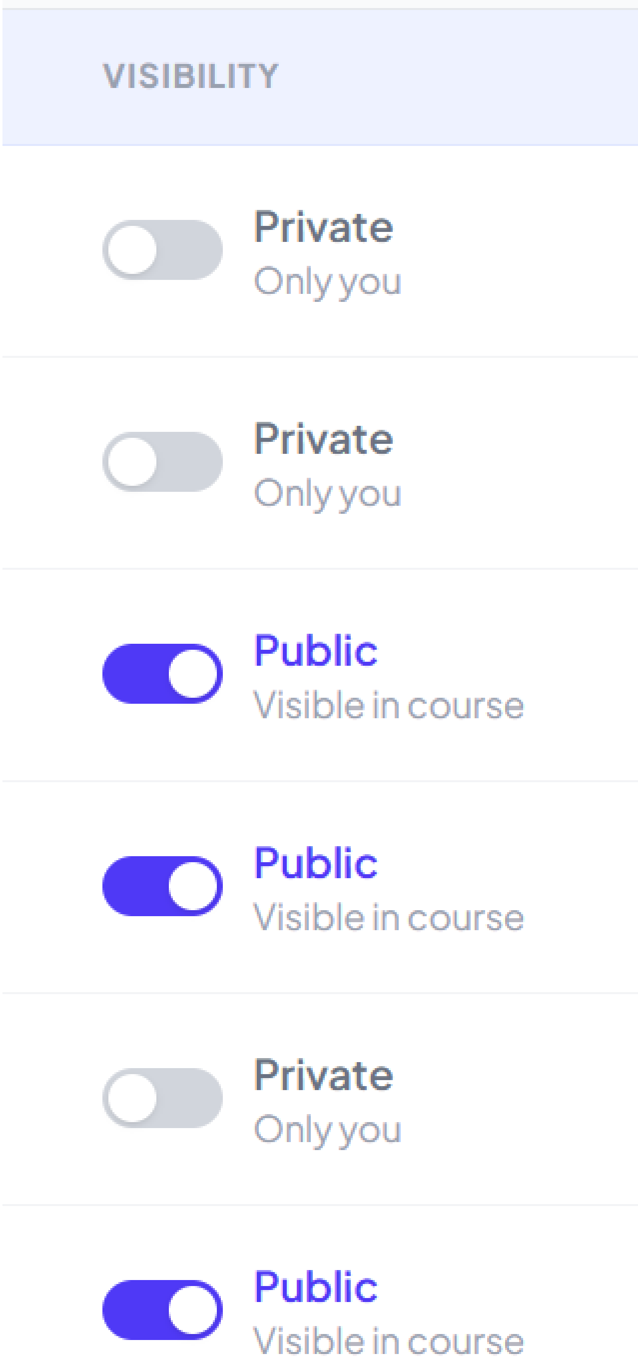


Figure 2.2.2 – Visibility toggle on a document row

- Visibility toggle (Private / Public).
- Visibility label (“Only you” / “Awaiting approval” / “Visible in course”).

2.5. Workflow: Read a Document (Document Viewer)

a. Purpose & Actor

- **Purpose:** Allow a student to read a document inline as a PDF, view its AI-generated summary, and download the original file. The same viewer is used for documents opened from the public library, the personal library, or Shared with me.
- **Actor:** Logged-in Student. Access is allowed when the viewer owns the document, the document is Public, or it has been shared with the viewer.

b. Step-by-Step Procedure

- **Step 1:** Open a document from any list (Home, a course, or Shared with me). The system opens the Document Viewer.
- **Step 2:** Read the document in the embedded PDF viewer. Use the built-in PDF controls to change page, zoom in/out, or print.
- **Step 3:** Read the AI Summary panel on the right (“Detailed Summary”). If summarization is still running, an in-progress indicator is shown until it completes.
- **Step 4:** Click the “Download” button (top-right) to save the original file.

Expected Result: The document is displayed with its AI summary, and the file can be downloaded.

c. Screenshots & UI References

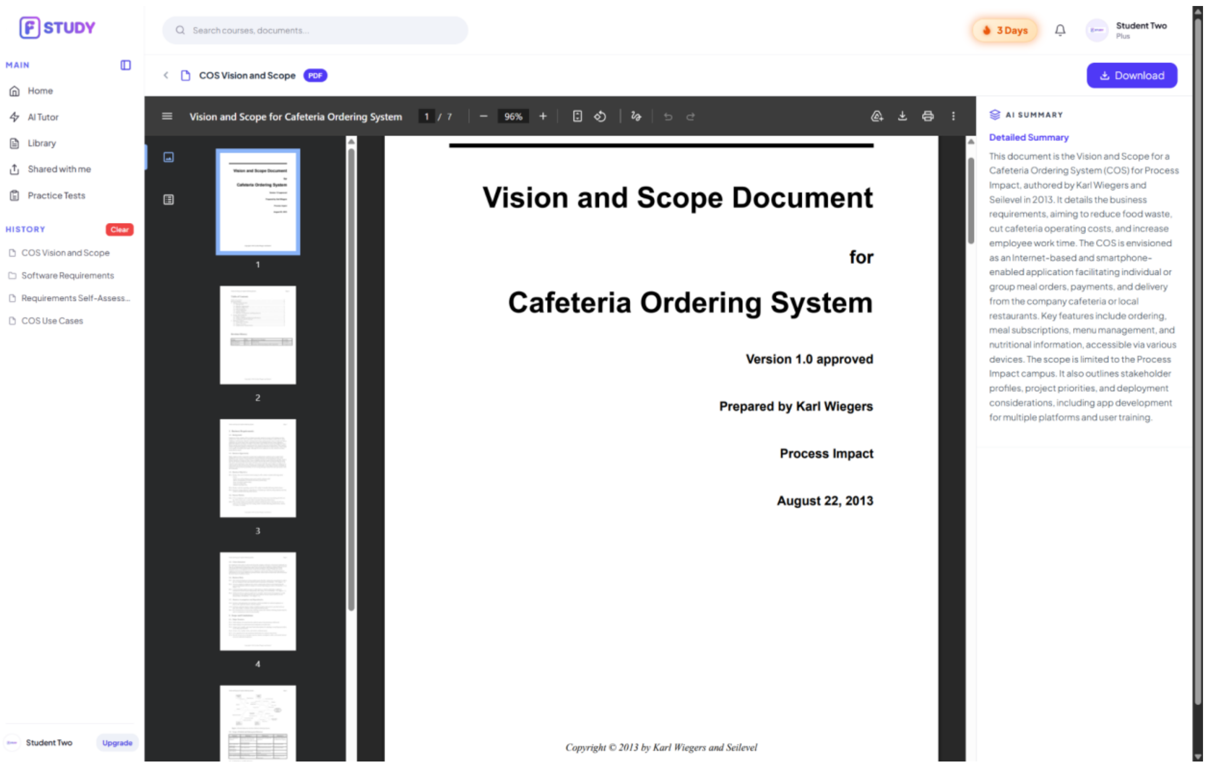


Figure 2.3.1 – Document Viewer

- (1) Document title with PDF badge.
- (2) “Download” button.
- (3) Embedded PDF viewer (page / zoom / print controls).
- (4) AI Summary panel (“Detailed Summary”).

2.6. Workflow: Course Management (Create & Delete Courses)

a. Purpose & Actor

- **Purpose:** Allow a student to organize their documents by course: create a new course in their personal library, and delete a course when it is no longer needed.
- **Actor:** Logged-in Student (owner of the personal library).

b. Step-by-Step Procedure

Part 1: Create a Course

- **Step 1:** Open Library from the sidebar. The page shows summary cards (Files, Storage Used, Courses, Generated) and the list of courses by semester.

- **Step 2:** Click the **“+ Create course”** button (top-right). The Create Course dialog opens.
- **Step 3:** Step – select semester: in the **“Select Semester”** step, choose the semester the course belongs to.
- **Step 4:** Step – select subject: search for and select the subject to add. Only subjects not already in your library are shown.
- **Step 5:** Confirm to create. The course is added to your library and appears in the list.

Expected Result: *The new course appears under its semester; documents can then be uploaded into it.*

Part 2: Delete a Course

- **Step 6:** On a course row, open the action menu (⁝) and click **Delete**.
- **Step 7:** Confirm in the “Delete ‘{course}’?” dialog.

Expected Result: *The course and all of its documents are removed. This is irreversible, so confirm carefully.*

Validation & Error Cases

- Duplicate course: adding a subject already in your library is rejected (already added).
- Empty search: if no subject matches the search term, the list shows no results.

c. Screenshots & UI References

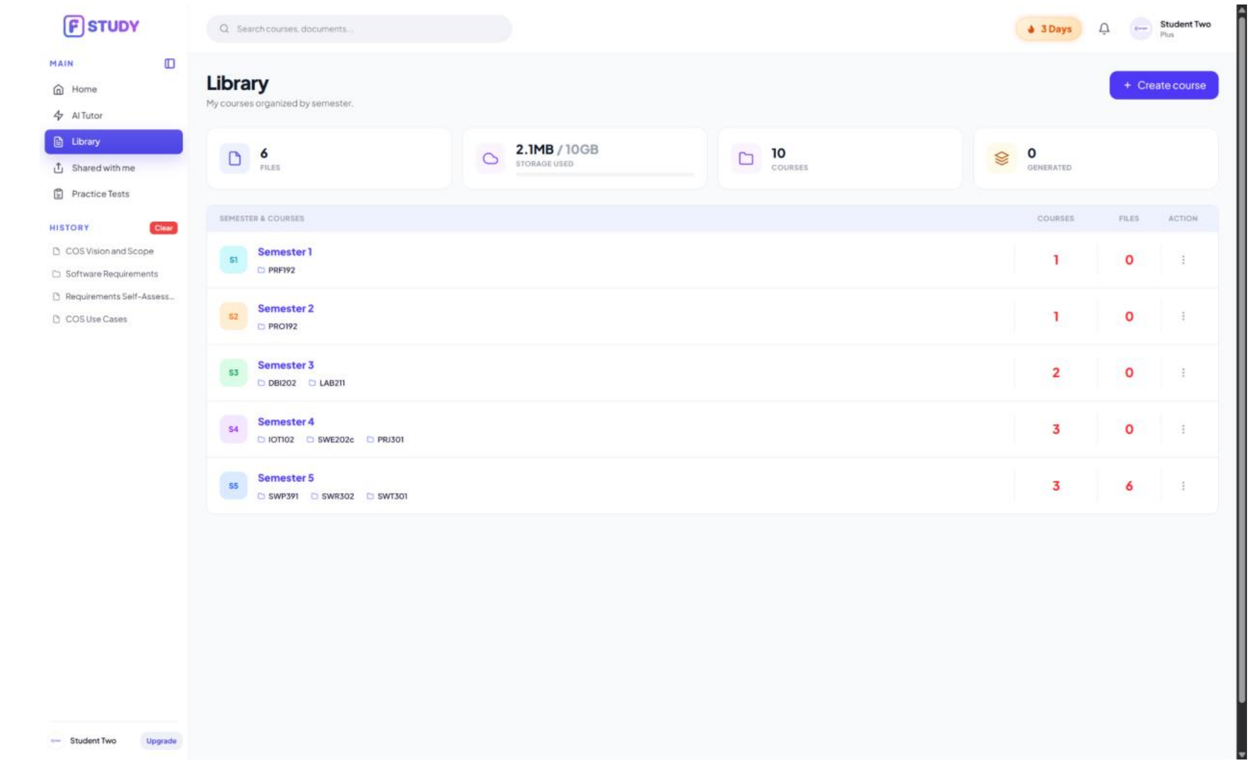


Figure 2.4.1 – Library page

- (1) Summary cards (Files, Storage Used, Courses, Generated).
- (2) “+ Create course” button.
- (3) Course row action menu (⁝) with Delete.

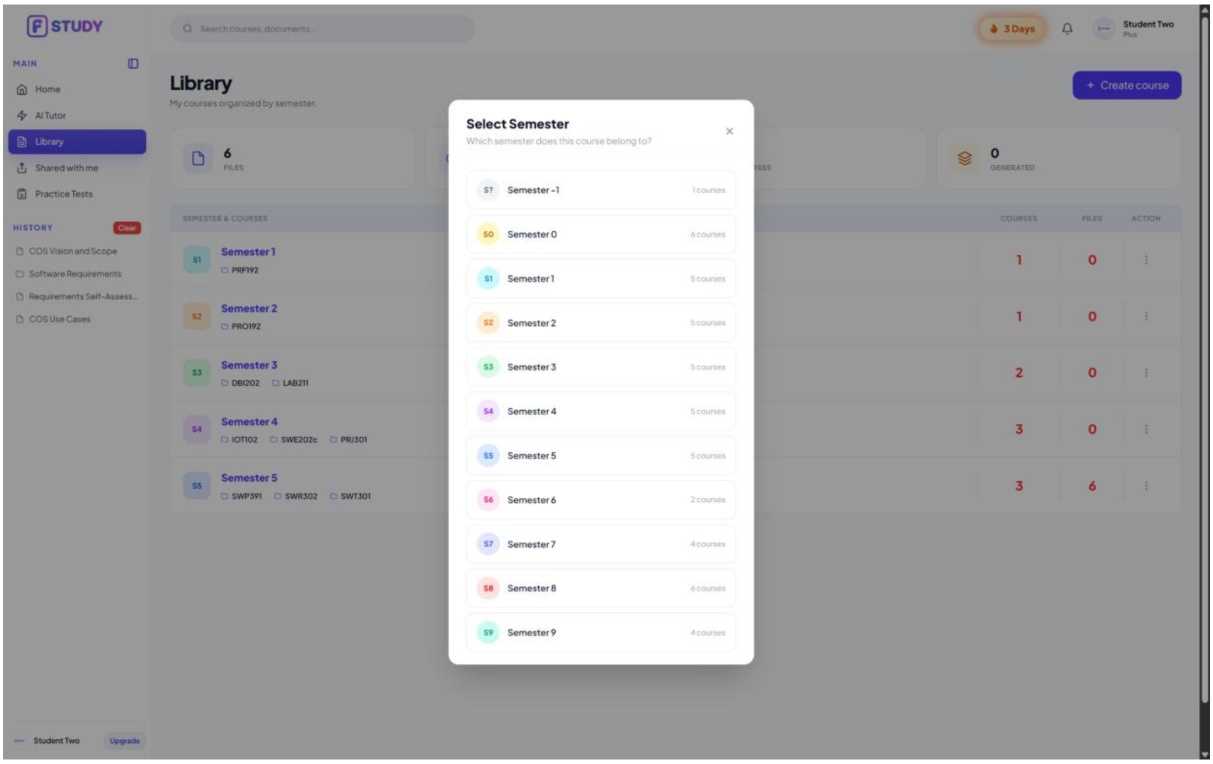


Figure 2.4.2 – Create Course dialog (Select Semester step)

- (1) Semester list (with course counts).
- (2) Subject search / selection step.

2.7. Workflow: Share a Document

a. Purpose & Actor

- **Purpose:** Allow the owner to share a document directly with specific users by email, granting each person a permission level (Viewer or Editor), and to manage who has access (change permission or remove access).
- **Actor:** Logged-in Student (document owner).

b. Step-by-Step Procedure

- **Step 1:** Open Library and go to the course that holds the document.
- **Step 2:** On the document row, open the action menu (**⋮**) and click **Share**. The Share dialog opens, titled "Share '{document}'".
- **Step 3:** In the **"Add people by email…"** field, enter the recipient's email address.
- **Step 4:** Choose the permission from the dropdown next to the field: **Viewer** (can read) or **Editor** (can edit).
- **Step 5:** Click the **"Share"** button. The recipient is added under "People with access" and the document appears in their "Shared with me".

Expected Result: *The recipient now has access to the document at the chosen permission level.*

Manage existing access

- In "People with access", each person has a dropdown to change their permission between Viewer and Editor.
- Select "Remove access" in that dropdown to revoke a person's access (this removes their access only; it does not delete the file).

Validation & Error Cases

- Empty email: clicking Share with no email shows “Please enter an email address.”
- Only the owner can share, change permissions, or remove access.

c. Screenshots & UI References

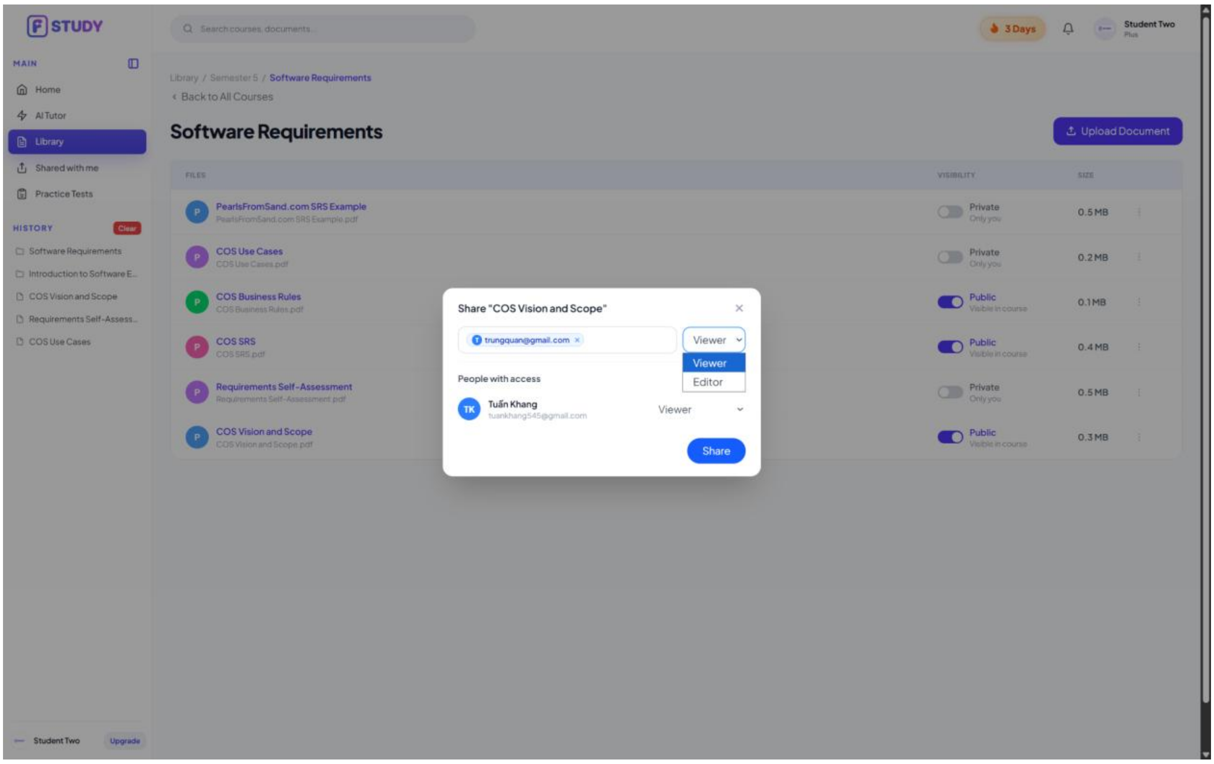


Figure 2.5.1 – Share dialog

- (1) “Add people by email...” field.
- (2) Permission dropdown (Viewer / Editor).
- (3) “Share” button.
- (4) “People with access” list, with per-person permission dropdown (Viewer / Editor / Remove access).

2.8. Workflow: AI Chat (Document Interaction) Workflow

a. Purpose & Actor

- **Purpose:** Allows logged-in students to ask questions about their study materials, get instant AI answers based on the uploaded documents with clear source links, search for files in their library, and ask follow-up questions easily without attaching the file again.
- **Actor:** Logged-in Student.

b. Step-by-Step Procedure

Part 1: Chatting with Selected Documents (Using the + Button)

- **Step 1:** On the top navigation bar or left sidebar, the student clicks the **"AI Chat"** tab to open the chat screen (/student/ai-chat).
- **Step 2:** The student clicks the **+ (Add icon)** button next to the chat text box. The system opens a popup menu showing options to choose study subjects or documents.
- **Step 3:** The student selects one or more existing documents (PDF, Word) or courses from their library. The selected items appear as small tags (chips) above the chat input box (e.g., [Doc: Chapter 1 Biology]).
- **Step 4:** The student types a study question into the input field (e.g., *"Summarize the main points of this document"*) and clicks the **"Send"** button (or presses Enter).
- **Step 5:** The system checks the document status in the database:

    - *If the file is still being processed by AI:* The system blocks the message and shows a warning notification: *"This document is still being processed. Please wait a moment."*
    - *If the file is ready:* The system searches the selected document to find the most relevant paragraphs.
- **Step 6:** The AI generates an accurate answer based only on the selected document. The chat box displays the answer formatted with bullet points, bold text, and clickable **Source Links** (e.g., [Source: Chapter 1 Biology]) at the bottom.
- **Step 7 (Checking the Source):** The student clicks on a source link. The system opens the PDF reader on the screen and automatically highlights the exact original text where the AI found the answer.

Part 2: Finding Documents & Asking Follow-up Questions

- **Step 8:** In a new chat message (without clicking the + button or attaching any file), the student types a search query directly into the chat box (e.g., *"Find study files about photosynthesis in my library"*) and clicks **"Send"**.
- **Step 9:** The AI searches through the student's entire document library in the database.
- **Step 10:** The system replies with a short summary and displays **Document Recommendation Cards** showing the best matching files found.
- **Step 11 (Follow-up Question):** The student continues the chat by asking a question about the recommended file (e.g., *"Can you explain the second diagram in this document?"*).
- **Step 12:** The system checks the recent chat history. When it detects words like *"this document"* or *"the file above"*, it automatically links the new question to the document found in **Step 10**.
- **Step 13:** The AI answers the follow-up question accurately using that document's content, saving the student from having to manually attach the file again. The system then saves the chat messages into the database.

c. Actual Screenshots & UI References

- **Screenshot 1 (Attaching a File):** Take a screenshot of the AI Chat page (/student/ai-chat) showing the **+ menu open**, with selected files displayed as blue tags above the input box. *(Draw a red circle around the + button and the attached file tags).*
- **Screenshot 2 (AI Answer with Source Links):** Take a screenshot of the chat screen showing the AI's reply, highlighting the clickable **Source Link tags** at the bottom of the message.
- **Screenshot 3 (Viewing the Original PDF):** Take a screenshot after clicking a source link, showing the chat box on one side and the PDF viewer on the other side with the referenced text highlighted in yellow.
- **Screenshot 4 (Document Search Result):** Take a screenshot showing the AI recommending study files in the chat when the user searches for a topic without attaching files.

3. Admin Workflows

3.1. Workflow A1: Master Data & Pricing Configuration Workflow

3.1.1 Task A1.1: Manage Subject Taxonomies & Course Categories

a. Purpose & Actor

- **Purpose:** Enables System Administrators to manage academic subject taxonomies, course categories, department codes, and credit units.
- **Actor:** System Administrator.

b. Step-by-Step Procedure

Part 1: Subject Catalog Exploration & Creation

- **Step 1:** On the Admin Sidebar, the admin selects 'Library Management' -> 'Subject Catalog'.
- **Step 2:** The system displays the Subject Management page with search filter and subject data table.

● **Step 3:** The admin clicks '+ Add New Subject' -> The system opens Create Subject dialog.

● **Step 4:** The admin enters Subject Code (e.g., PRJ301), Title, Department, and Credit Units, then clicks 'Save Subject'.

● **Step 5:** The system checks for duplicate subject codes:

○ ○ If code exists: The system blocks creation with error: 'Subject code PRJ301 already exists in system.'

○ ○ If valid: The system creates subject record and displays notification: 'Subject created successfully.'

Part 2: Subject Editing & Safe Deletion Validation

● **Step 6:** The admin locates subject in data table and clicks 'Edit' icon.

● **Step 7:** The system opens Edit Subject dialog pre-filled with current subject data.

● **Step 8:** The admin modifies title or credit units and clicks 'Save Changes'.

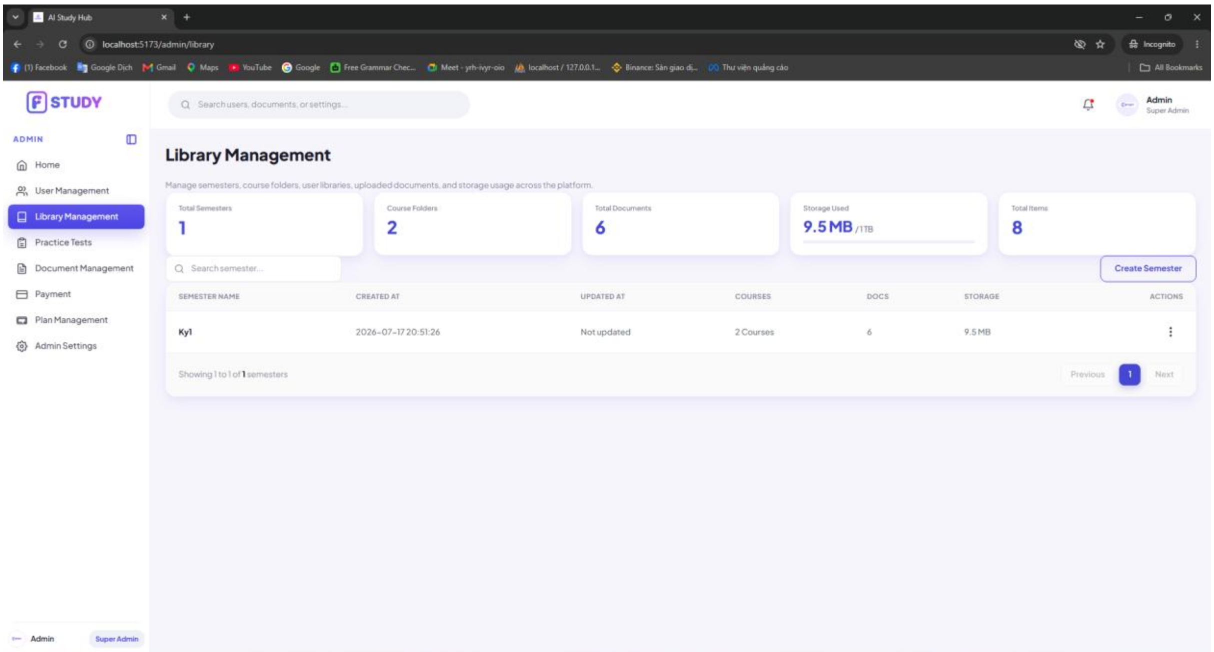
● **Step 9:** The admin clicks 'Delete' icon on a subject row.

● **Step 10:** The system checks for assigned documents:

○ ○ If active documents assigned: The system rejects deletion with message: 'Cannot delete subject because documents are currently assigned to it.'

○ ○ If no documents assigned: The system removes subject from database.

c. Actual Screenshots & UI References



3.1.2 Task A1.2: Manage Add-on AI Features & Storage Packs

a. Purpose & Actor

- **Purpose:** Allows administrators to configure add-on packs (Extra Vector Storage, Fast AI Speed) available for student purchase.
- **Actor:** System Administrator.

b. Step-by-Step Procedure

Part 1: Add-on Configuration & Publishing

● **Step 1:** Admin navigates to 'Pricing Management' -> 'Add-on Services' tab.

● **Step 2:** The system displays active add-on features and price cards.

● **Step 3:** Admin clicks '+ Create Add-on' -> System opens Create Add-on dialog modal.

● **Step 4:** Admin inputs Name, Category, Price (VND), and Quota Value, then clicks 'Save Add-on'.

● **Step 5:** The system saves add-on record and publishes it to student pricing checkout.

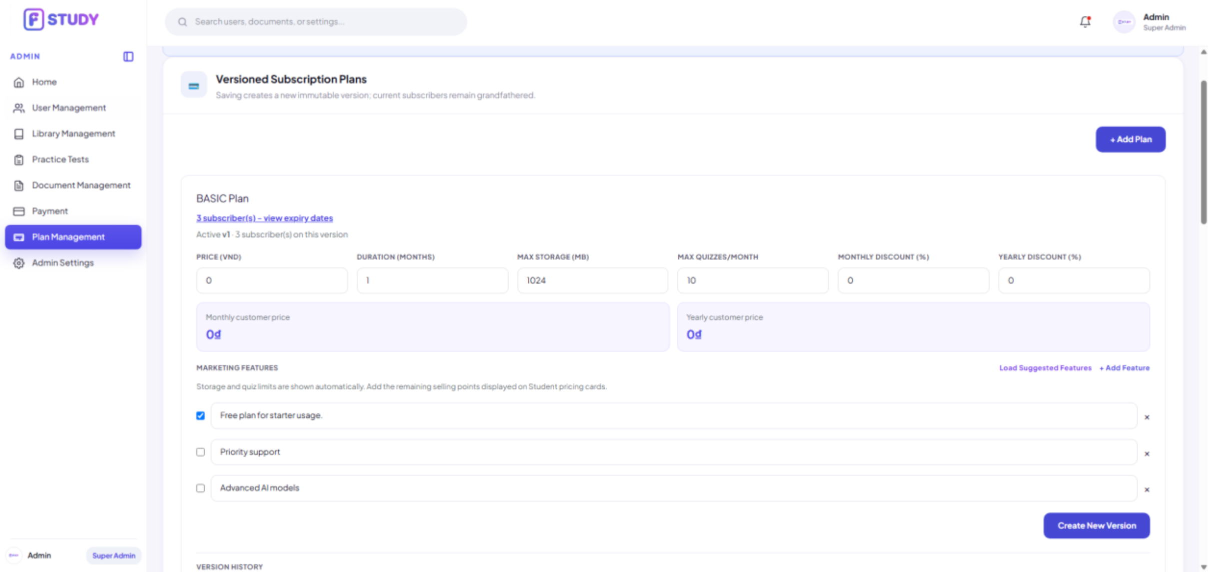
Part 2: Add-on Modification & Removal

● **Step 6:** Admin clicks 'Edit' on add-on card -> Modifies pricing or quota -> Clicks 'Save Changes'.

● **Step 7:** Admin clicks 'Delete' icon on add-on card -> System displays confirmation modal.

● **Step 8:** Admin confirms deletion -> System removes add-on feature from store.

c. Actual Screenshots & UI References



3.1.3 Task A1.3: Manage Individual Student Subscription Plans

a. Purpose & Actor

- **Purpose:** Enables System Administrators to configure tier parameters for individual student subscription plans (Free, Pro, Premium).
- **Actor:** System Administrator.

b. Step-by-Step Procedure

Part 1: Subscription Tier Overview & Editing

● **Step 1:** Admin navigates to 'Pricing Management' -> 'Student Plans'.

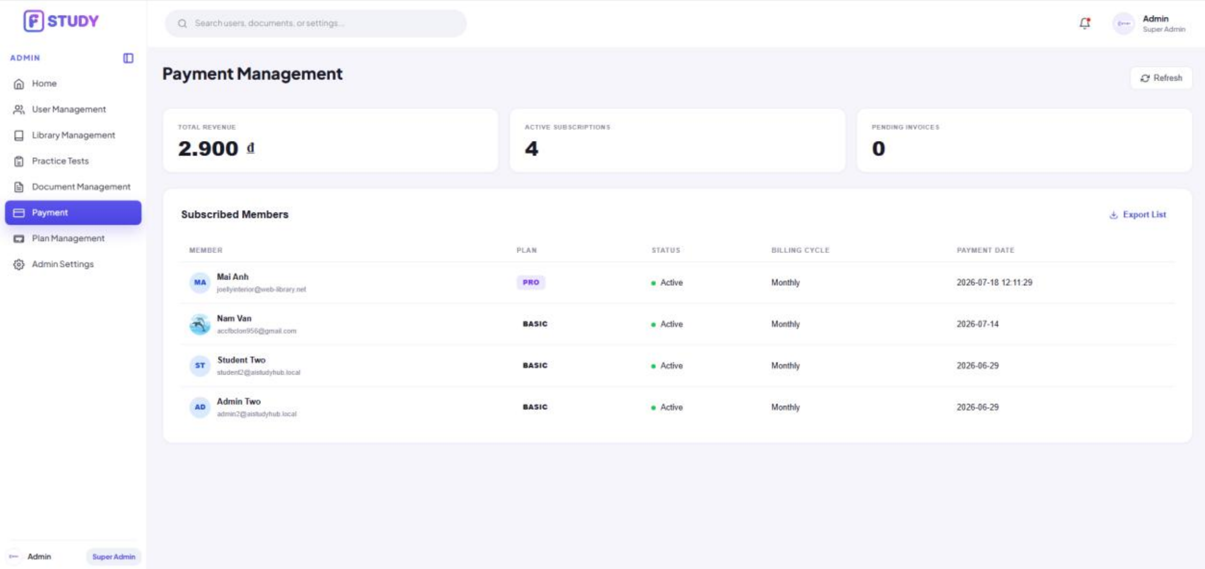
● **Step 2:** The system displays plan cards (Free, Pro, Premium) showing monthly fees and quota caps.

● **Step 3:** Admin clicks 'Edit Plan' on Pro Tier card.

● **Step 4:** Admin modifies Monthly Fee, Daily Query Limit, or Upload Cap, then clicks 'Save Changes'.

● **Step 5:** The system updates plan parameters in database and refreshes student pricing table.

c. Screenshots



3.1.4 Task A1.4: Manage Group & University License Subscription Plans

a. Purpose & Actor

- **Purpose:** Allows admins to create, edit, and delete Group/Classroom and University-wide license subscription plans.
- **Actor:** System Administrator.

b. Step-by-Step Procedure

Part 1: Institutional License Plan Management

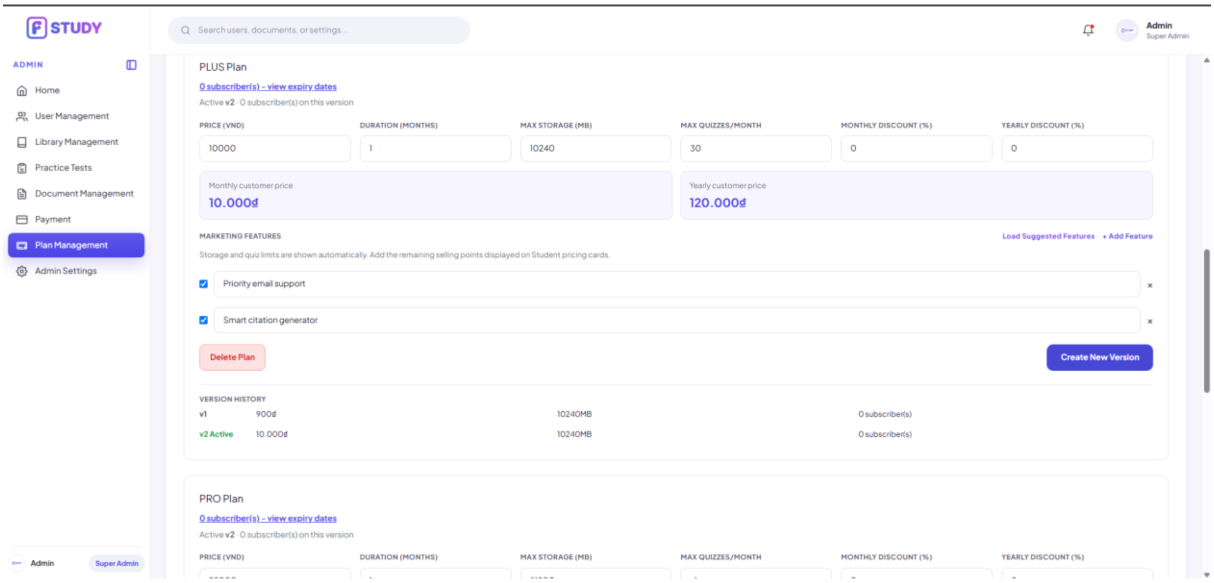
● **Step 1:** Admin navigates to 'Pricing Management' -> 'Group & University Plans'.

● **Step 2:** Admin clicks '+ Create Group Plan' -> System opens form dialog.

● **Step 3:** Admin enters Plan Name (Campus Study Group 10-Pack), Max Member Seats (10), Pooled Storage, and Price.

● **Step 4:** Admin clicks 'Save Plan' -> System persists institutional subscription plan.

c. Screenshots



3.2. Workflow A2: Employee, Moderator & Student Management Workflow

a. Purpose & Actor

- **Purpose:** Enables System Administrators to manage student accounts, adjust AI quotas, manage content moderator accounts, and assign administrative roles.
- **Actor:** System Administrator.

b. Step-by-Step Procedure

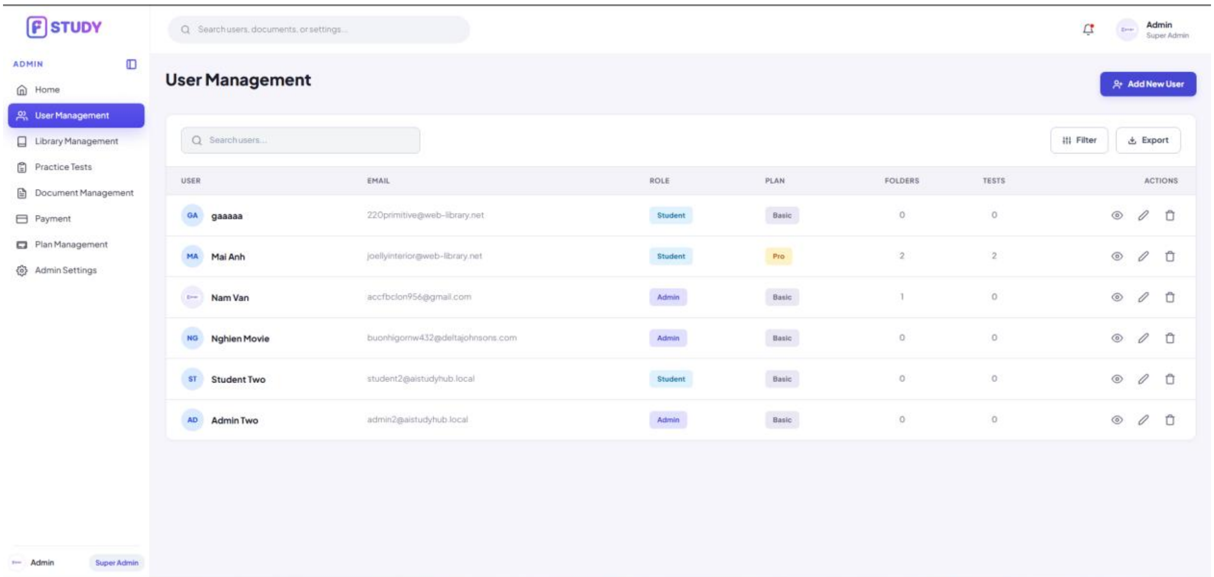
Part 1: Student Account Moderation & Quota Adjustments

- **Step 1:** Admin navigates to 'User Management' -> 'Student Directory'.
- **Step 2:** The system displays student list with search filter and status badges.
- **Step 3:** Admin searches student by email and clicks row to open Student Drawer.
- **Step 4:** Admin inspects upload history, adjusts quota overrides, or toggles 'Suspend Account'.
- **Step 5:** The system updates user status: If suspended, active JWT sessions are invalidated immediately.

Part 2: Staff & Content Moderator Onboarding

- **Step 6:** Admin switches to 'Staff & Moderator Management' tab and clicks '+ Add Moderator'.
- **Step 7:** Admin enters Name, Email, Department, and assigns Role (Content Moderator).
- **Step 8:** Admin clicks 'Create Account' -> System sends invitation email with temp password.

c. Screenshots



3.3. Workflow A3: Practice Test Management Workflow

a. Purpose & Actor

- **Purpose:** Enables System Administrators to monitor and manage AI-generated practice tests, review test questions and correct answers, evaluate test statistics, edit question content, and remove inappropriate or unnecessary practice tests.
- **Actor:** System Administrator.

b. Step-by-Step Procedure

Part 1: Practice Test Monitoring & Management

- **Step 1:** Admin navigates to **“Practice Tests”** from the Admin sidebar.

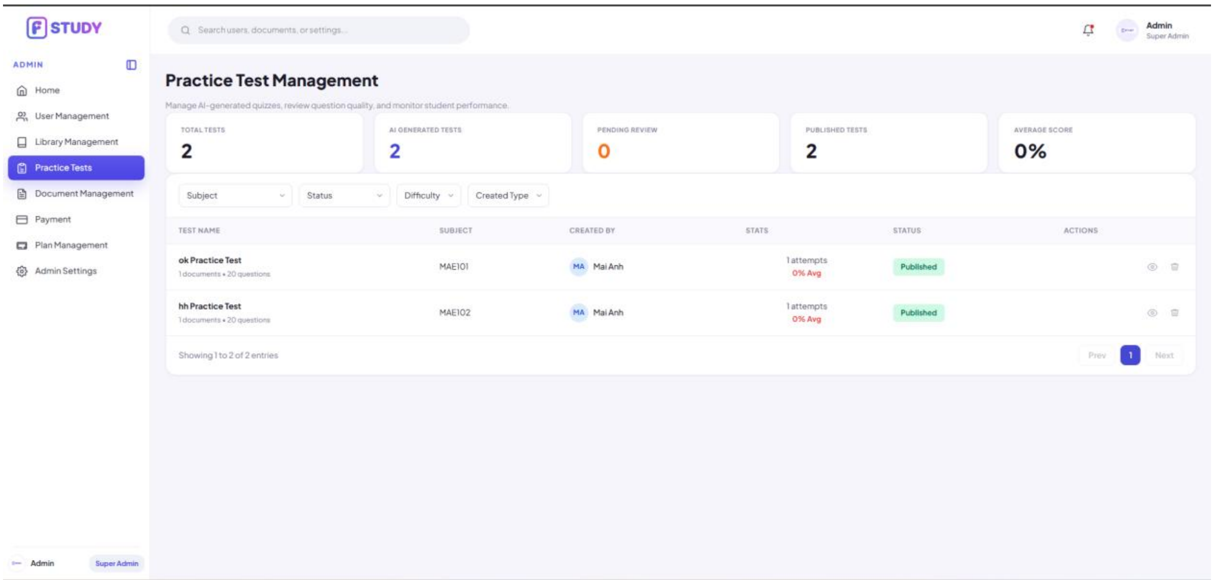
- **Step 2:** The system displays the **Practice Test Management** page, including summary information:
  - Total Tests
  - AI Generated Tests
  - Pending Review
  - Published Tests
  - Average Score
- **Step 3:** Admin uses the available filters to search for practice tests by:
  - Subject
  - Status
  - Difficulty
  - Created Type
- **Step 4:** The system displays the practice test list with the following information:
  - Test Name
  - Subject
  - Created By
  - Number of Attempts
  - Average Score
  - Status
  - Available Actions
- **Step 5:** Admin reviews the number of student attempts and average score to evaluate the usage and performance of each practice test.
- **Step 6:** Admin clicks the **View icon** in the Actions column to open the selected practice test details.
- **Step 7:** Admin may click the **Delete icon** to remove an unnecessary or inappropriate practice test.
- **Step 8:** Before deleting the test, the system displays a confirmation message to prevent accidental deletion.
- **Step 9:** After Admin confirms the deletion, the system removes the practice test and updates the test list and summary statistics.

Part 2: Practice Test Question Review & Editing

- **Step 10:** On the Practice Test Detail page, the system displays general test information, including:
  - Test Name
  - Subject
  - Creator
  - Publication Status
  - Total Questions
  - Total Attempts
  - Average Score
  - Number of Documents Used
- **Step 11:** The system displays the complete question list of the selected practice test.
- **Step 12:** For each question, the system shows:
  - Question Number
  - Question Type
  - Difficulty Level
  - Question Content
  - Answer Options
  - Correct Answer

- **Step 13:** The correct answer is highlighted so that Admin can quickly verify the accuracy of the AI-generated question.
- **Step 14:** Admin reviews the question content, answer options, difficulty level, and correct answer.
- **Step 15:** Admin clicks the **Edit icon** next to a question that contains incorrect, unclear, duplicated, or inappropriate content.
- **Step 16:** The system opens the question editing form with the current question information.
- **Step 17:** Admin updates the necessary information, such as:
    - Question Content
    - Answer Options
    - Correct Answer
    - Question Type
    - Difficulty Level
- **Step 18:** Admin clicks **"Save"** to submit the changes.
- **Step 19:** The system validates the updated information and displays an error message if:
    - The question content is empty.
    - An answer option is missing.
    - No correct answer is selected.
    - Multiple-choice answers are duplicated.
    - The submitted data exceeds the allowed length.
- **Step 20:** If the information is valid, the system saves the changes and displays the updated question in the question list.
- **Step 21:** Admin clicks **"Back to Practice Tests"** to return to the Practice Test Management page.

c. Screenshots



3.4. Workflow A4: Document Management Workflow

a. Purpose & Actor

- **Purpose:** Enables System Administrators to review uploaded documents, monitor document status, preview document content, verify uploaded information, and remove inappropriate or unnecessary documents before they are available to students.
- **Actor:** System Administrator.

b. Step-by-Step Procedure

Part 1: Document Review & Management

- Step 1: Admin navigates to "Document Management" from the Admin sidebar.
- Step 2: The system displays the Document Management dashboard, including summary statistics:
  - Total Documents
  - Pending Review
  - Approved Documents
  - Rejected Documents
  - Total Storage Size
- Step 3: Admin filters documents using:
  - Course
  - Status
  - Document Type
- Step 4: The system displays the document list, including:
  - Document Name
  - Course
  - Uploaded By
  - Upload Time
  - File Size
  - Review Status
  - Available Actions
- Step 5: Admin reviews document information and verifies whether the uploaded file meets the platform's requirements.
- Step 6: Admin clicks the View icon in the Actions column to open the selected document.
- Step 7: Admin may click the Delete icon to permanently remove documents that violate policies, contain incorrect content, or are no longer required.
- Step 8: Before deleting the document, the system displays a confirmation dialog to prevent accidental deletion.
- Step 9: After confirmation, the system deletes the document and automatically updates the document list, storage statistics, and document counters.

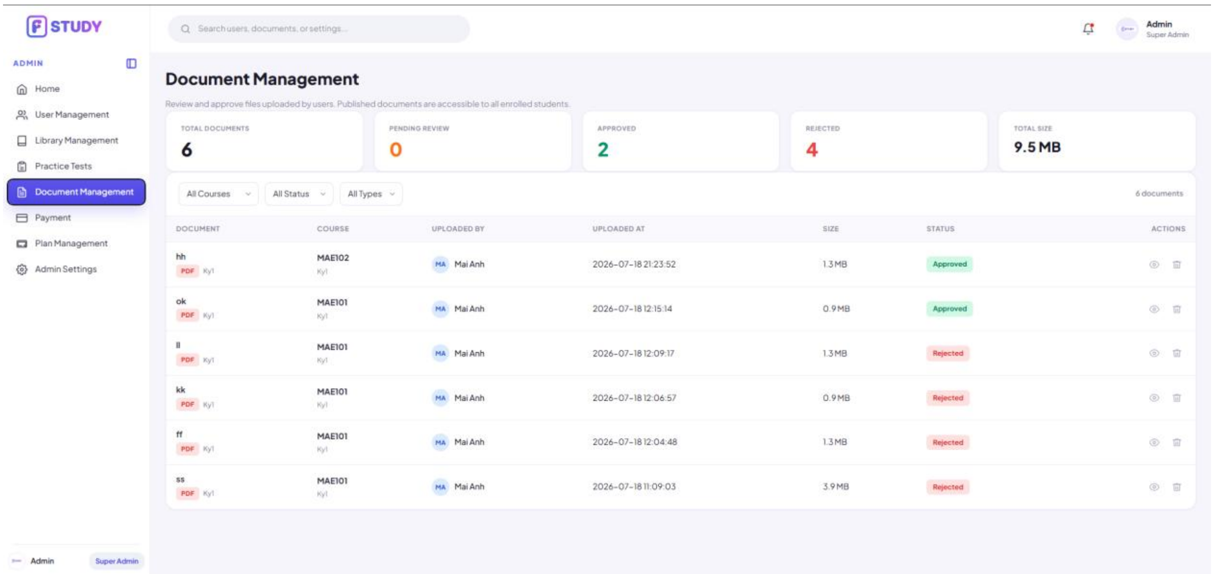
Part 2: Document Preview & Verification

- Step 10: The system opens the Document Preview window.
- Step 11: The built-in PDF viewer loads the selected document and displays page thumbnails for quick navigation.
- Step 12: Admin reviews the document content page by page using the integrated PDF viewer.
- Step 13: The system displays the document metadata, including:
  - Uploaded By
  - Upload Time
  - Current Status
  - Description
  - Download Button
- Step 14: Admin verifies:
  - Document content
  - File completeness

- Upload information
- File size
- Course association
- Current approval status

- Step 15: If necessary, Admin clicks Download to download the original document for further inspection.
- Step 16: After reviewing the document, Admin closes the preview window and returns to the Document Management page.

c. Screenshots



3.5. Workflow A5: Admin Settings Workflow

a. Purpose & Actor

- **Purpose:** Enables System Administrators to configure platform-wide settings, including document storage providers, AI service configuration, API keys, AI model parameters, and system-wide notifications for users.
- **Actor:** System Administrator.

b. Step-by-Step Procedure

Part 1: Document Storage Configuration

- Step 1: Admin navigates to "Admin Settings" from the Admin sidebar.
- Step 2: The system displays the Admin Settings page.
- Step 3: Admin opens the Document Storage section.
- Step 4: The system displays the available storage providers:
  - Supabase Storage
  - Cloudflare R2
- Step 5: Admin selects the preferred storage provider for newly uploaded documents.
- Step 6: The system highlights the selected storage provider while informing that existing files remain in their current location.
- Step 7: Admin clicks "Save Storage Setting."
- Step 8: The system validates the configuration and saves the selected storage provider successfully.

Part 2: Send Notification to Users

- Step 9: Admin opens the Send Notification to Users section.
- Step 10: Admin selects the notification recipients (e.g., All Users).
- Step 11: Admin chooses the notification type:
    - Info
    - Announcement
    - Warning
    - Alert
- Step 12: Admin enters the notification title.
- Step 13: Admin writes the notification message.
- Step 14: Admin clicks "Send Notification."
- Step 15: The system validates the required fields and broadcasts the notification to the selected recipients.
- Step 16: After successful delivery, the notification is recorded in the Recent Notifications Sent history.
- Step 17: Admin may edit or delete previously sent notifications from the notification history.

Part 3: AI API & Model Configuration

- Step 18: Admin opens the AI API & Model Configuration section.
- Step 19: The system displays the available AI providers, including:
    - OpenAI
    - Google Gemini
    - DeepSeek
- Step 20: Admin selects the preferred AI provider.
- Step 21: Admin enters or replaces the API Key.
- Step 22: Admin clicks "Test Connection" to verify the API configuration.
- Step 23: The system validates the API Key and displays the connection status.
- Step 24: Admin selects the AI model (e.g., GPT-4o Mini, GPT-4o, GPT-4.1 Mini, GPT-4.1).
- Step 25: Admin configures the AI generation parameters, including:
    - Temperature
    - Max Tokens
    - Top P
- Step 26: Admin clicks "Save AI Configuration."
- Step 27: The system validates all configuration values and securely stores the updated AI settings.

c. Screenshots